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# Grapevine Cultivar Discrimination Using Proximal Hyperspectral Data and Machine Learning

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## ACRONYMS AND ABBREVIATIONS

|     |                                   |
|-----|-----------------------------------|
| ASD | Analytical Spectral Devices       |
| AUC | Area Under the Curve              |
| DNA | Deoxyribonucleic Acid             |
| FN  | False Negative                    |
| FP  | False Positive                    |
| MCC | Matthews Correlation Coefficient  |
| nm  | Nanometre                         |
| OA  | Overall Accuracy                  |
| OvR | One-vs-Rest                       |
| PA  | Producer's Accuracy               |
| PCA | Principal Component Analysis      |
| PLS | Partial Least Squares             |
| RF  | Random Forest                     |
| RFE | Recursive Feature Elimination     |
| SVM | Support Vector Machine            |
| TN  | True Negative                     |
| UA  | User's Accuracy                   |
| VIP | Variable Importance in Projection |

## 1. Introduction

Grapevine cultivar discrimination is an important problem in vineyard management because accurate identification of cultivars supports site-specific decisions related to irrigation, nutrients, pest management, inventory capture, and harvest estimation. Accurate varietal identification is important during the growing season because different cultivars have unique management requirements and varietal information contributes to viticultural inventory and supply planning. Destructive methods such as wet chemistry, DNA-based sequencing, or other laboratory analyses are time-consuming, labour-intensive, expensive, and impractical for routine operational classification (Gutiérrez et al., 2018), whereas non-destructive spectral acquisition offers a rapid alternative (Mirzaei et al., 2019).

Proximal hyperspectral sensing is well suited to cultivar discrimination because it measures reflectance continuously across hundreds of narrow, contiguous wavebands. Hyperspectral remote sensing is characterised by high spectral resolution and narrow contiguous wavebands, which makes it especially useful for detailed discrimination of heterogeneous objects (Thenkabail et al., 2000). However, high-dimensional datasets often contain large numbers of redundant predictors, and performance can deteriorate when the number of features becomes too large relative to the number of training samples a phenomenon known as the "Hughes effect" or the curse of dimensionality. Loggenberg et al. (2026) says that hyperspectral datasets can capture subtle cultivar differences but can also reduce classifier generalisability if informative spectral subsets are not identified.

Machine learning can be used with hyperspectral data for varietal discrimination. Mirzaei et al. (2019) describe field spectroscopy as an accurate, rapid, and non-destructive technique for grapevine variety identification and report strong test accuracies using SVM. Furthermore, Karakizi et al. (2016) shows that variety discrimination is complex due to the high spectral similarity between different vine types, necessitating robust classification frameworks. This study focuses on six raisin grapevine cultivars: Currants, Merbein Seedless, Diamond Muscat, Selma Pete, Sugra-39, and Sultana sampled at the Welgevallen Experimental Farm in Stellenbosch.

## 1.1 RESEARCH PROBLEM

The research problem in this study was how to develop a theoretically sound and technically correct machine-learning framework for discriminating six raisin grapevine cultivars from proximal hyperspectral ASD leaf spectra, while accounting for high dimensionality, mild class imbalance, and temporal instability between acquisition dates.

This problem had several linked dimensions. First, the spectral data were high-dimensional. After removal of atmospheric absorption windows, 1,874 wavebands remained. In most cases dimensionality reduction is necessary because it reduces multicollinearity, storage demands, and computation time, and may improve model accuracy. There are two main reduction strategies: feature selection, which preserves original predictors, and feature extraction, which transforms the original data into a reduced space, for example using PCA.

Second, the study had to address the problem of model choice. In this study, Random Forest served as the ensemble learner, while Support Vector Machine served as the non-ensemble learner. Random Forest is widely regarded as robust to high-dimensional data and can help mitigate the Hughes effect. SVM, by contrast, is a strong classifier for hyperspectral data because it constructs an optimal separating hyperplane and is effective in high-dimensional feature spaces. Loggenberg et al. (2026) describe SVM as a linear classifier that maximises the margin between classes, while Mirzaei et al.(2019) explain that SVM maps data into higher-dimensional feature space and constructs an optimal separating hyperplane for class discrimination.

Third, the problem included the selection and evaluation of dimensionality-reduction methods. Feature-selection methods are not universally optimal across learners. Loggenberg et al.(2026) state that feature-selection subsets are rarely optimal across different classifiers and that wrapper methods often outperform learner-independent filter methods, particularly when classification performance depends strongly on the learner-feature interaction. This made it important to test classifier-specific feature-selection methods rather than assume that one reduction strategy would suit all models equally well.

Fourth, the study had to address class imbalance. Although the December and January datasets were only mildly imbalanced, the minority class Currants had fewer samples than the other cultivars. Imbalanced training data can lead to class bias and poor test accuracy, and recommends the use of appropriate metrics such as precision, recall, F1-score.

Finally, the study had to address temporal transferability. The problem was not only whether cultivars could be discriminated accurately in December, but whether the best-performing model remained effective when transferred to January. Loggenberg et al. (2026) shows that cultivar discrimination is



complicated by temporal variability related to phenological stage, management, and environmental conditions.

## **1.2 AIMS AND OBJECTIVES**

The aim of this study was to develop and evaluate a machine-learning framework for discriminating six raisin grapevine cultivars from proximal hyperspectral leaf spectra collected in Stellenbosch.

The objectives of the study were:

1. To preprocess raw ASD spectral measurements to construct cleaned, modeling-ready datasets for December 2023 and January 2024
2. To remove known atmospheric absorption windows and construct cleaned sample and waveband datasets suitable for modelling.
3. To assess class balance and account for mild class imbalance using both model-level and evaluation-level approaches.
4. To compare the performance of two classifiers on the December dataset, namely Random Forest as the ensemble learner and Support Vector Machine as the non-ensemble learner.
5. To evaluate at least one feature-selection method per classifier.
6. To refine the best-performing December model using hyperparameter optimisation.
7. To assess feature transformation of the refined best-model workflow using PCA and spectral binning.
8. To evaluate temporal transferability by applying the best-performing model workflow to the January 2024 dataset.
9. To interpret the findings in relation to hyperspectral theory, machine-learning principles, and the relevant grapevine discrimination literature.

### 1.3 RESEARCH METHODOLOGY AND DESIGN

Two raw ASD datasets were used. December 2023 was the primary modelling dataset, while January 2024 was reserved for temporal transferability assessment. The raw ASD data were first cleaned by removing atmospheric absorption windows at 350–399 nm, 1350–1425 nm, 1825–1925 nm, and 2451–2500 nm. This left 1,874 retained wavebands for modelling.

The December dataset was then split into training and test subsets using a 70/30 stratified split. This was methodologically appropriate because it preserved class proportions while ensuring that final model comparison was based on previously unseen December test data. Importantly, all preprocessing steps that needed fitting from the data such as scaling, Boruta selection, RFE selection, VIP selection, grid search, and PCA were fitted using the December training data only. January was not used in training, tuning, scaling, feature selection, or PCA fitting.

Class imbalance was handled using two approaches. The first was model-level handling through class weight="balanced" in Random Forest and SVM. The second was evaluation-level handling using Balanced Accuracy, Macro F1, MCC, and Macro AUC. Balanced Accuracy and F1-score are used for imbalanced classification. Balanced Accuracy is preferable to simple accuracy for imbalanced classes, and that MCC is especially useful because it only becomes high when performance is good across all entries of the confusion matrix.

The classification framework comprised six models. Model 1 was a baseline Random Forest using all retained wavebands. Model 2 was a baseline linear SVM using all retained wavebands. Model 3 applied Boruta wrapper feature selection before Random Forest. Model 4 applied Recursive Feature Elimination before SVM. Model 5 applied PLS-VIP feature selection before SVM. Model 6 then refined the best earlier model. And the best model is used to test temporal transferability. Figure 1 below illustrates the methodology.

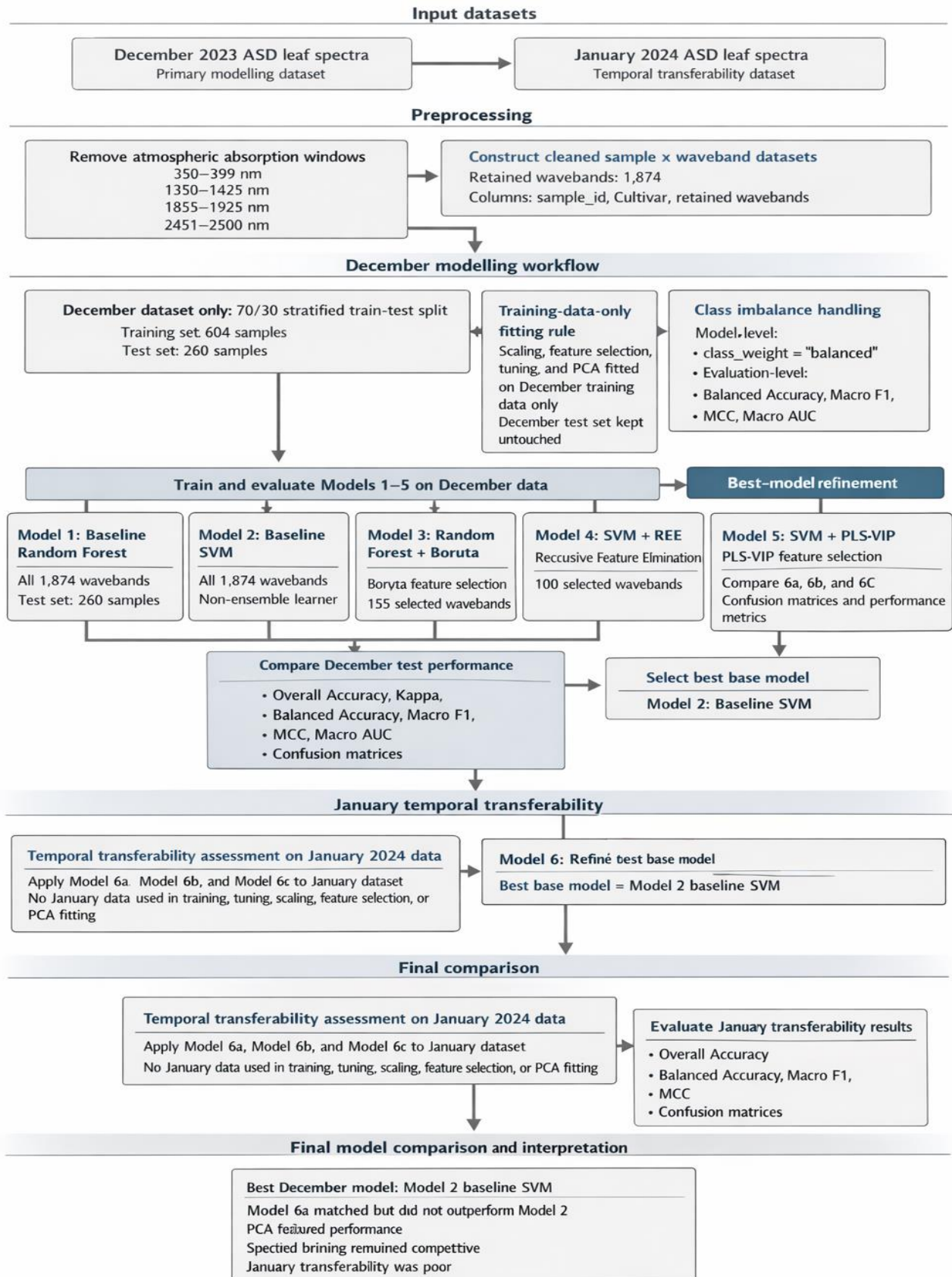


Figure 1: Flow Diagram

## 2. Methods

### 2.1 Study Area

The study used proximal hyperspectral data collected at the Welgevallen Experimental Farm in Stellenbosch, South Africa. The data was collected between December 2023 until January 2024.



Figure 2: Study Area

### 2.2 Datasets and preprocessing

The study started with raw proximal hyperspectral reflectance data collected in December 2023 ( $n = 864$ ) and January 2024 ( $n = 852$ ). The raw spectra were captured across 2,151 contiguous wavebands ranging from 350 nm to 2500 nm.

All the preprocessing, modelling and evaluation was done in VS Code in Python. To remove spectral noise associated with atmospheric water absorption, the following wavelength regions were excluded, wavebands 350–399 nm, 1350–1425 nm, 1825–1925 nm, and 2451–2500 nm. The resulting cleaned dataset consisted of 864 samples and 1874 spectral features for the December

dataset. And 852 samples and 1874 spectral features for the January dataset. Table 1 illustrates the number of samples captured per class for each dataset.

| Cultivar Type    | December Count | January Count |
|------------------|----------------|---------------|
| Merbein Seedless | 153            | 153           |
| Selma Pete       | 153            | 153           |
| Sultana          | 153            | 141           |
| Diamond Muscat   | 144            | 144           |
| Sugra-39         | 144            | 144           |
| Currants         | 117            | 117           |
| <b>Total</b>     | <b>864</b>     | <b>852</b>    |

Table 1: Class Distribution Table

## 2.3 Exploratory data inspection

Initial inspection focused on class distribution, sample numbers, and the overall behaviour of the cleaned spectra, this was done to understand the class structures and to look for things like spectral overlap. Spectral overlap was expected, because grapevine cultivar discrimination is difficult precisely due to high intra-class variability and low inter-class differences, as noted by Loggenberg et al.(2026).

## 2.4 Class imbalance handling

Imbalanced data can cause class bias and poor test performance, and this was mitigated through proper metrics and other strategies. In this study, two required imbalance approaches were applied. First, class weights were used at model level through `class weight="balanced"` in RF and SVM. Second, evaluation-level mitigation was achieved through the reporting of Balanced Accuracy, Macro F1, MCC, and Macro AUC, rather than relying only on Overall Accuracy. This was especially important because the dataset was mildly imbalanced and because the minority class Currants still needed to be classified reliably.

The December dataset was split using a 70/30 stratified split, yielding 604 training samples and 260 test samples. Stratification preserved class proportions but was treated as good practice rather than as one of the two required imbalance approaches.

## 2.5 Classification framework

Random Forest was selected as the ensemble learner and Support Vector Machine as the non-ensemble learner. Both algorithms are commonly used in high-dimensional classification, but they represent fundamentally different learning principles.

### 2.5.1 Model 1: Random Forest

Random Forest is an ensemble classifier that builds many decision trees on bootstrapped subsets of the data and combines their outputs through voting. It is effective in high-dimensional settings because it can model non-linear relationships and is relatively robust to multicollinearity and noisy predictors. Random Forest performs well at mitigating the Hughes effect and is robust towards high dimensionality.

The Random Forest model was configured with 500 trees ( $n_{\text{estimators}}=500$ ) to ensure ensemble stability and a robust majority vote, as smaller ensembles may fail to capture the complexity of high-dimensional spectral data. To mitigate the 'curse of dimensionality,' max features was set to the square root of the total features ('sqrt'), a standard heuristic for classification that forces tree diversity by considering only a subset of wavebands at each split.

Node quality was measured using the Gini impurity ( $\text{criterion}='gini'$ ), and trees were allowed to grow to their full depth ( $\text{max depth}=\text{None}$ ) to maximize local discriminative power. To address the inherent class imbalance, the  $\text{class weight}='balanced'$  parameter was applied, which automatically adjusts weights inversely proportional to class frequencies in the training data. Finally, the Out-of-Bag ( $\text{oob score}=\text{True}$ ) setting was enabled to provide an internal cross-validation estimate, ensuring the model's generalization capability was monitored during the ensemble construction.

### 2.5.2 Model 2: SVM

Support Vector Machine is a margin-based classifier. Loggenberg et al. (2026) describe SVM as a linear classifier that constructs an optimal separating hyperplane by maximising the margin between classes. Gutiérrez et al. (2018) found that linear kernels produce the highest SVM scores among tested kernels. These properties make SVM especially appropriate for high-dimensional hyperspectral data where many correlated variables may still yield good class separation in margin-based space.

The Support Vector Machine was implemented using a linear kernel, which is ideal for high-dimensional datasets where the number of features ( $p = 1874$ ) greatly exceeds the number of samples ( $n = 604$  in training). In such 'large  $p$ , small  $n$  scenarios', class boundaries are often linearly separable in the high-dimensional feature space, making complex non-linear kernels unnecessary and prone to overfitting (Gutiérrez et al., 2018).

The regularisation parameter was set to  $C=1.0$ , providing a standard balance between maximizing the margin width and minimizing training misclassifications. To compensate for class imbalance, `class_weight="balanced"` was employed. This ensures that the hyperplane is not biased toward majority cultivars (like Sultana) and instead penalizes misclassifications of the minority class (Currants) more heavily during the optimization process.

The decision function `shape="ovr"` (One-vs-Rest) approach was selected to handle the multiclass nature of the six raisin cultivars, while `probability=True` was enabled to allow for the calculation of the Macro AUC (Area Under the Curve). Finally, the model was wrapped in a `Standard Scaler()` pipeline. This was essential because SVM is a distance-based algorithm; without standardizing the wavebands to a mean of zero and a unit variance, features with higher absolute reflectance values would disproportionately dominate the margin calculation, leading to an inaccurate decision boundary.

### **2.5.3 Model 3: Random Forest + Boruta**

Boruta is a wrapper method around Random Forest. The Boruta Wrapper method is an all-relevant wrapper algorithm embedded with RF that creates shuffled shadow features, compares original features against the most important shadow features, and iteratively confirms, rejects, or leaves features tentative based on importance scores. Boruta is useful where the aim is to retain all relevant information, not just a minimal predictor subset.

Model 3 tested whether RF performance improved after removing redundant or irrelevant wavebands using Boruta. It selected 159 wavebands, with 404 nm, 405 nm, and 411 nm among the top-ranked bands.

### **2.5.4 Model 4: SVM + RFE**

Recursive Feature Elimination is a wrapper-based feature-selection method that recursively removes less important predictors and refits the learner until a smaller subset remains. Because RFE is learner-dependent, it was a suitable choice for SVM, where the aim was to retain the most discriminative spectral subset for the classifier rather than preserve all potentially relevant bands. In this study the model selected 100 wavebands, with 400nm, 404nm, and 405 nm as the highest ranked features.

### 2.5.5 Model 5: SVM + PLS-VIP

PLS-based feature selection was included because PLS is widely used with high-dimensional spectral data. Loggenberg et al. (2026) explain that PLS is a foundational framework for cultivar discrimination because of its effectiveness with high-dimensional datasets, while Mirzaei et al. (2019) report that PLS-based approaches can identify informative wavelengths for grapevine variety discrimination.

The PLS-VIP procedure used 15 latent components (max components=15) to maximize the covariance between the spectra and cultivar labels while avoiding noise. A VIP threshold of 1.0 was applied to filter out non-influential wavebands. Retaining the top 100 bands provided a minimalist feature subset, focusing primarily on the 400–410 nm (blue) region, which is biologically linked to variations in leaf pigment concentration.

## 2.6 Best-model refinement

The best-performing model, model 2 the SVM model was improved using hyperparameter optimisation and feature transformation.

### 2.6.1 Model 6a: SVM + Grid Search

To optimise the best-performing model, GridSearchCV was employed using Balanced Accuracy as the primary tuning score. This ensured the optimization process prioritized performance across all six cultivars equally, specifically protecting the accuracy of the minority class (Currants). The search space evaluated the trade-off between a Linear kernel and a Non-linear (RBF) kernel across varying levels of regularization ( $C = 0.1, 1, 10, 100$ ). The selection of a linear kernel with  $C=1$  validates that the initial baseline was mathematically near optimal for this dataset. In high-dimensional spectral problems where features ( $p=1874$ ) far outnumber samples ( $n=604$ ), a linear margin is often superior because it provides a stable decision boundary that resists overfitting to high-frequency noise (Gutiérrez et al., 2018). The  $C=1$  result indicates an ideal balance between maximizing the margin width and minimizing misclassification, ensuring robust generalisation to unseen December test data.

### 2.6.2 Model 6b: SVM + Grid Search + PCA

PCA was implemented to transform the 1,874 wavebands into a reduced set of orthogonal components that summarize global variance rather than preserving original variables. Setting components=0.95 ensured that 95% of the spectral information was retained while eliminating redundant dimensions.



### 2.6.3 Model 6c: SVM + Grid Search + Spectral binning

Spectral binning aggregated neighbouring bands into 5 nm groups. Thenkabail et al. (2000) argue that the aggregation of narrow, contiguous wavebands can enhance the signal-to-noise ratio and provide a more robust representation of agricultural crop characteristics by reducing redundant spectral dimensions Loggenberg et al. (2026).

## 2.7 Accuracy Assessment

Thematic metrics reported in this study were Overall Accuracy, Kappa, Producer's Accuracy, User's Accuracy, Omission Error, and Commission Error. Producer's accuracy is how many reference samples of a class were correctly classified on the map, and user's accuracy is how many samples assigned to a class on the classified map are that class. Overall Accuracy summarises the proportion of correctly mapped reference sites, and Kappa improves on OA by accounting for chance agreement.

Machine-learning metrics reported were Balanced Accuracy, Macro Precision, Macro Recall, Macro F1, MCC, and Macro AUC (OvR). Precision is equivalent to user's accuracy, Recall is equivalent to producer's accuracy, F1 is the harmonic mean of precision and recall, and Balanced Accuracy is used for imbalanced data. Multiclass AUC is calculated in a one-vs-rest framework and that MCC is particularly strong for imbalanced classification because it reflects performance across TP, FP, TN, and FN. Appendix A shows a detailed summary result of the metrics used in the accuracy assessment.

## 2.8 Temporal transferability

Temporal transferability was tested using the best-performing model workflow (Model 6a). To ensure a nonbiased assessment, the January 2024 dataset was treated as a completely independent validation set. No January samples were used during the training, feature selection, or hyperparameter tuning phases; instead, the model trained solely on December data and was applied to the January samples to determine its stability across different phenological stages

### 3. Results

#### 3.1 Dataset overview and class balance

The cleaning and window-removal process successfully reduced the spectral feature space from 2,151 to 1,874 wavebands while preserving the 864 December samples. This removal was necessary to eliminate atmospheric noise that could otherwise degrade classifier generalisation. This confirms that the modelling problem remained strongly high-dimensional even after atmospheric filtering. The December and January class distributions were only mildly imbalanced, but Currants was consistently the minority class, which justified the use of class weighting and imbalance-aware metrics. Figure 3 shows the data distribution after cleaning the data.

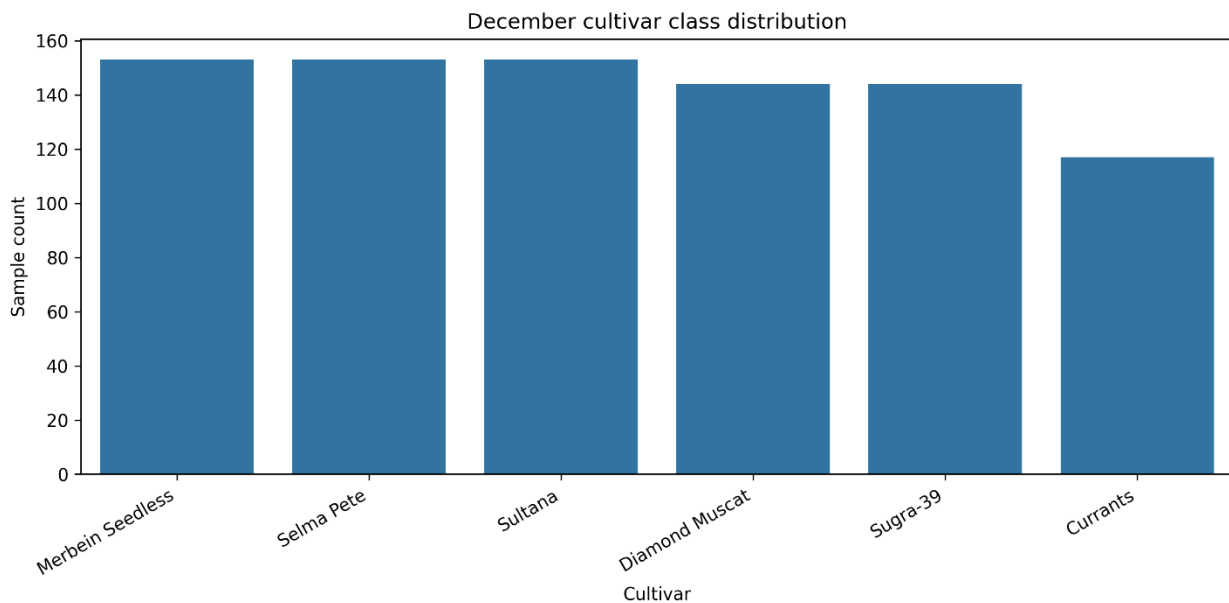


Figure 3: December cultivar class distribution

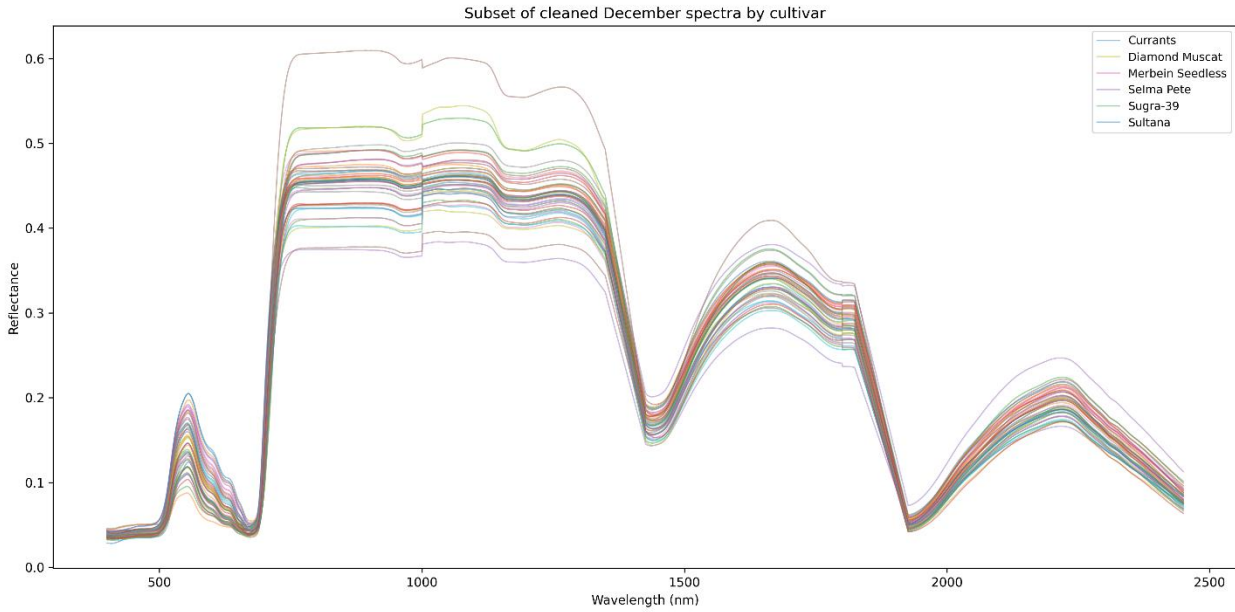


Figure 4: Subset of cleaned December Spectra by Cultivar

The spectral behaviour of the cultivars showed broad overlap, which is consistent with the grapevine literature. Loggenberg et al.(2026) noted that cultivar mapping is challenging because of high intra-class heterogeneity and low inter-class variability. That difficulty helps explain why model selection and dimensionality reduction were central components of this study rather than optional refinements

### 3.2 Model comparison

A comprehensive comparison of all global classification metrics (OA, Kappa, MCC, Balanced Accuracy, and Macro F1) across the nine experimental models is provided in Appendix A.

### 3.2.1 Model 1: Random Forest

Model 1 achieved OA = 0.5538, Kappa = 0.4634, MCC = 0.4641, and Macro F1 = 0.5585 on the December test set. This can be seen in figure 5 below which shows the confusion matrix. RF is robust to high dimensionality; it performed much worse than SVM. The model exhibited significant overfitting, as evidenced by perfect training metrics that failed to generalize to unseen test data. These results highlight that theoretical suitability does not guarantee optimal performance in subtle cultivar-discrimination tasks; consequently, while RF satisfied the ensemble-learning requirement, it was not the superior learner for this dataset compared to the SVM.

| CLASSIFIED DATA               | REFERENCE DATA      |                |                  |            |          |         |        | COMMISSIONS | USER'S ACCURACY |
|-------------------------------|---------------------|----------------|------------------|------------|----------|---------|--------|-------------|-----------------|
|                               | Currants            | Diamond Muscat | Merbein Seedless | Selma Pete | Sugra-39 | Sultana | TOTALS |             |                 |
|                               | Currants            | 26             | 2                | 1          | 3        | 3       | 0      | 35          | 74,29           |
|                               | Diamond Muscat      | 0              | 17               | 5          | 7        | 11      | 4      | 44          | 38,64           |
|                               | Merbein Seedless    | 0              | 4                | 18         | 12       | 2       | 10     | 46          | 39,13           |
|                               | Selma Pete          | 2              | 7                | 6          | 26       | 1       | 4      | 46          | 56,52           |
|                               | Sugra-39            | 4              | 6                | 4          | 1        | 24      | 4      | 43          | 55,81           |
|                               | Sultana             | 0              | 3                | 6          | 1        | 3       | 33     | 46          | 71,74           |
|                               | TOTALS              | 32             | 39               | 40         | 50       | 44      | 55     | 260         |                 |
|                               | OMISSIONS           | 6              | 22               | 22         | 24       | 20      | 22     |             |                 |
|                               | PRODUCER'S ACCURACY | 81,25          | 43,59            | 45,00      | 52,00    | 54,55   | 60,00  |             |                 |
| K                             | 0,1231              | 0,1500         | 0,1538           | 0,1923     | 0,1692   | 0,2115  |        |             |                 |
| % Cover                       | 12,31               | 15,00          | 15,38            | 19,23      | 16,92    | 21,15   |        |             |                 |
| Total correct classifications | 144                 |                |                  |            |          |         |        |             |                 |
| Total number of samples       | 260                 |                |                  |            |          |         |        |             |                 |
| Kappa                         | 0,463               |                |                  |            |          |         |        |             |                 |
| Overall accuracy              | 0,554               |                |                  |            |          |         |        |             |                 |

Figure 5: Confusion Matrix for RF Model

### 3.2.2 Model 2: SVM

Model 2 (Baseline SVM) was the study's top-performing December model, achieving an OA of 0.8923, an MCC of 0.8709, and a Macro AUC of 0.9829. This can be seen in figure below which shows the confusion matrix Its success over Random Forest is attributed to the maximum-margin hyperplane, which is better suited for high-dimensional spectral data than recursive partitioning (Gutiérrez et al., 2018; Mirzaei et al., 2019).

Thematic results were robust across the class set, showing no bias toward majority classes. Despite its small sample size, Currants achieved a producer's accuracy of 88.57% and a user's accuracy of 93.94%. Peak class performances were observed for Sultana (97.83% producer's accuracy) and Selma Pete (95.12% user's accuracy), confirming the model's high reliability for specific varietal discrimination.

|                 | REFERENCE DATA                |                |                  |            |          |         |        | COMMISSIONS | USER'S ACCURACY |
|-----------------|-------------------------------|----------------|------------------|------------|----------|---------|--------|-------------|-----------------|
|                 | Currants                      | Diamond Muscat | Merbein Seedless | Selma Pete | Sugra-39 | Sultana | TOTALS |             |                 |
| CLASSIFIED DATA | Currants                      | 31             | 1                | 1          | 1        | 0       | 35     | 4           | 88,57           |
|                 | Diamond Muscat                | 0              | 40               | 1          | 0        | 3       | 44     | 4           | 90,91           |
|                 | Merbein Seedless              | 1              | 0                | 40         | 0        | 1       | 46     | 6           | 86,96           |
|                 | Selma Pete                    | 1              | 0                | 3          | 39       | 2       | 46     | 7           | 84,78           |
|                 | Sugra-39                      | 0              | 3                | 3          | 0        | 37      | 43     | 6           | 86,05           |
|                 | Sultana                       | 0              | 0                | 0          | 1        | 0       | 45     | 46          | 97,83           |
|                 | TOTALS                        | 33             | 44               | 48         | 41       | 44      | 50     | 260         |                 |
|                 | OMISSIONS                     | 2              | 4                | 8          | 2        | 7       | 5      |             |                 |
|                 | PRODUCER'S ACCURACY           | 93,94          | 90,91            | 83,33      | 95,12    | 84,09   | 90,00  |             |                 |
|                 | K                             | 0,1269         | 0,1692           | 0,1846     | 0,1577   | 0,1692  | 0,1923 |             |                 |
|                 | % Cover                       | 12,69          | 16,92            | 18,46      | 15,77    | 16,92   | 19,23  |             |                 |
|                 | Total correct classifications | 232            |                  |            |          |         |        |             |                 |
|                 | Total number of samples       | 260            |                  |            |          |         |        |             |                 |
|                 | Kappa                         | 0,871          |                  |            |          |         |        |             |                 |
|                 | Overall accuracy              | 0,892          |                  |            |          |         |        |             |                 |

Figure 6: Confusion Matrix For SVM Model

### 3.2.3 Model 3: Random Forest + Boruta

Model 3 (RF + Boruta) achieved an OA of 0.5500 and an MCC of 0.4597. This can be seen in figure 7 below which shows the confusion matrix. Despite reducing the feature space to 159 wavebands, performance did not improve meaningfully over the baseline. This happened, because Boruta identifies informative features but often retains correlated predictors, which fails to resolve the underlying redundancy issues impacting the Random Forest's generalisation.

| CLASSIFIED DATA | REFERENCE DATA                |          |                |                  |            |          |         | COMMISSIONS | USER'S ACCURACY |
|-----------------|-------------------------------|----------|----------------|------------------|------------|----------|---------|-------------|-----------------|
|                 |                               | Currants | Diamond Muscat | Merbein Seedless | Selma Pete | Sugra-39 | Sultana |             |                 |
|                 | Currants                      | 28       | 0              | 2                | 3          | 2        | 0       | 35          | 7               |
|                 | Diamond Muscat                | 0        | 18             | 6                | 7          | 11       | 2       | 44          | 26              |
|                 | Merbein Seedless              | 1        | 4              | 19               | 12         | 3        | 7       | 46          | 27              |
|                 | Selma Pete                    | 2        | 8              | 4                | 28         | 0        | 4       | 46          | 18              |
|                 | Sugra-39                      | 4        | 6              | 3                | 3          | 23       | 4       | 43          | 20              |
|                 | Sultana                       | 0        | 3              | 9                | 4          | 3        | 27      | 46          | 19              |
|                 | TOTALS                        | 35       | 39             | 43               | 57         | 42       | 44      | 260         |                 |
|                 | OMISSIONS                     | 7        | 21             | 24               | 29         | 19       | 17      |             |                 |
|                 | PRODUCER'S ACCURACY           | 80,00    | 46,15          | 44,19            | 49,12      | 54,76    | 61,36   |             |                 |
|                 | K                             | 0,1346   | 0,1500         | 0,1654           | 0,2192     | 0,1615   | 0,1692  |             |                 |
|                 | % Cover                       | 13,46    | 15,00          | 16,54            | 21,92      | 16,15    | 16,92   |             |                 |
|                 | Total correct classifications |          |                |                  |            |          |         |             |                 |
|                 | Total number of samples       | 143      |                |                  |            |          |         |             |                 |
|                 |                               | 260      |                |                  |            |          |         |             |                 |
|                 | Kappa                         | 0,459    |                |                  |            |          |         |             |                 |
|                 | Overall accuracy              | 0,550    |                |                  |            |          |         |             |                 |
|                 |                               |          |                |                  |            |          |         |             |                 |
|                 |                               |          |                |                  |            |          |         |             |                 |

Figure 7: Confusion Matrix For RF + Boruta Model

### 3.2.4 Model 4: SVM +RFE

Model 4 (SVM + RFE) achieved an OA of 0.8769 and an MCC of 0.8530, maintaining high accuracy while reducing the feature space from 1,874 to just 100 wavebands. This outcome confirms the efficacy of learner-aligned wrapper methods, proving that a minimalist hyperspectral classifier remains highly competitive when the retained wavebands are truly discriminative.

| CLASSIFIED DATA               | REFERENCE DATA   |         |                |                  |            |          |         |        |             |                 |
|-------------------------------|------------------|---------|----------------|------------------|------------|----------|---------|--------|-------------|-----------------|
|                               |                  | Curants | Diamond Muscat | Merbein Seedless | Selma Pete | Sugra-39 | Sultana | TOTALS | COMMISSIONS | USER'S ACCURACY |
|                               | Curants          | 33      | 2              | 0                | 0          | 0        | 0       | 35     | 2           | 94,29           |
|                               | Diamond Muscat   | 0       | 41             | 0                | 1          | 2        | 0       | 44     | 3           | 93,18           |
|                               | Merbein Seedless | 0       | 3              | 36               | 0          | 0        | 7       | 46     | 10          | 78,26           |
|                               | Selma Pete       | 0       | 0              | 1                | 43         | 1        | 1       | 46     | 3           | 93,48           |
|                               | Sugra-39         | 1       | 5              | 3                | 0          | 34       | 0       | 43     | 9           | 79,07           |
|                               | Sultana          | 0       | 1              | 1                | 2          | 1        | 41      | 46     | 5           | 89,13           |
|                               | TOTALS           | 34      | 52             | 41               | 46         | 38       | 49      | 260    |             |                 |
|                               | OMISSIONS        | 1       | 11             | 5                | 3          | 4        | 8       |        |             |                 |
| PRODUCER'S ACCURACY           | 97,06            | 78,85   | 87,80          | 93,48            | 89,47      | 83,67    |         |        |             |                 |
|                               |                  |         |                |                  |            |          |         |        |             |                 |
| K                             | 0,1308           | 0,2000  | 0,1577         | 0,1769           | 0,1462     | 0,1885   |         |        |             |                 |
| % Cover                       | 13,08            | 20,00   | 15,77          | 17,69            | 14,62      | 18,85    |         |        |             |                 |
|                               |                  |         |                |                  |            |          |         |        |             |                 |
| Total correct classifications | 228              |         |                |                  |            |          |         |        |             |                 |
| Total number of samples       |                  |         |                |                  |            |          |         |        |             |                 |
|                               |                  | 260     |                |                  |            |          |         |        |             |                 |
|                               |                  |         |                |                  |            |          |         |        |             |                 |
| Kappa                         | 0,852            |         |                |                  |            |          |         |        |             |                 |
| Overall accuracy              | 0,877            |         |                |                  |            |          |         |        |             |                 |

Figure 8: Confusion Matrix for SVM + RFE model

### 3.2.5 Model 5:SVM + PLS-VIP

Model 5 (SVM + PLS-VIP) achieved an OA of 0.7154 and an MCC of 0.6588, underperforming relative to both the baseline and RFE-optimised SVM. Although PLS-based feature selection is robust for high-dimensional spectral data this result highlights that feature subsets are learner-dependent and not universally optimal.

| CLASSIFIED DATA               | REFERENCE DATA   |          |                |                  |            |          |         | COMMISSIONS | USER'S ACCURACY |        |    |       |
|-------------------------------|------------------|----------|----------------|------------------|------------|----------|---------|-------------|-----------------|--------|----|-------|
|                               |                  | Currants | Diamond Muscat | Merbein Seedless | Selma Pete | Sugra-39 | Sultana |             |                 | TOTALS |    |       |
|                               | Currants         | 33       | 2              | 0                | 0          | 0        | 0       |             |                 | 35     | 2  | 94,29 |
|                               | Diamond Muscat   | 1        | 28             | 2                | 1          | 10       | 2       |             |                 | 44     | 16 | 63,64 |
|                               | Merbein Seedless | 1        | 5              | 26               | 7          | 2        | 5       |             |                 | 46     | 20 | 56,52 |
|                               | Selma Pete       | 1        | 1              | 7                | 33         | 1        | 3       |             |                 | 46     | 13 | 71,74 |
|                               | Sugra-39         | 2        | 10             | 1                | 0          | 30       | 0       |             |                 | 43     | 13 | 69,77 |
|                               | Sultana          | 0        | 2              | 1                | 6          | 1        | 36      |             |                 | 46     | 10 | 78,26 |
|                               | TOTALS           | 38       | 48             | 37               | 47         | 44       | 46      |             |                 | 260    |    |       |
|                               | OMISSIONS        | 5        | 20             | 11               | 14         | 14       | 10      |             |                 |        |    |       |
| PRODUCER'S ACCURACY           | 86,84            | 58,33    | 70,27          | 70,21            | 68,18      | 78,26    |         |             |                 |        |    |       |
| K                             | 0,1462           | 0,1846   | 0,1423         | 0,1808           | 0,1692     | 0,1769   |         |             |                 |        |    |       |
| % Cover                       | 14,62            | 18,46    | 14,23          | 18,08            | 16,92      | 17,69    |         |             |                 |        |    |       |
| Total correct classifications | 186              |          |                |                  |            |          |         |             |                 |        |    |       |
| Total number of samples       |                  |          |                |                  |            |          |         |             |                 |        |    |       |
|                               |                  | 260      |                |                  |            |          |         |             |                 |        |    |       |
| Kappa                         | 0,658            |          |                |                  |            |          |         |             |                 |        |    |       |
| Overall accuracy              | 0,715            |          |                |                  |            |          |         |             |                 |        |    |       |

Figure 9: Confusion Matrix for SVM + PLS-VIP Model

## 3.3 Best-model Refinement

After evaluating the 5 models, model 2 (SVM) was the best performing model. Model 2 was refined using hyperparameter optimisation and using feature transformation.



### 3.3.1 Model 6a: SVM + Grid Search

Model 6a (Tuned SVM) achieved an OA of 0.8923 and an MCC of 0.8709, exactly matching the performance Model 2. GridSearchCV confirmed that the original parameters (kernel='linear', C=1) were already mathematically optimal for the December dataset. This model served as a statistical validation of the baseline model rather than future performance gains.

| CLASSIFIED DATA               | REFERENCE DATA   |          |                |                  |            |          |         | COMMISSIONS | USER'S ACCURACY |        |
|-------------------------------|------------------|----------|----------------|------------------|------------|----------|---------|-------------|-----------------|--------|
|                               |                  | Currants | Diamond Muscat | Merbein Seedless | Selma Pete | Sugra-39 | Sultana |             |                 | TOTALS |
|                               | Currants         | 31       | 1              | 1                | 1          | 1        | 0       | 35          | 4               | 88,57  |
|                               | Diamond Muscat   | 0        | 40             | 1                | 0          | 3        | 0       | 44          | 4               | 90,91  |
|                               | Merbein Seedless | 1        | 0              | 40               | 0          | 1        | 4       | 46          | 6               | 86,96  |
|                               | Selma Pete       | 1        | 0              | 3                | 39         | 2        | 1       | 46          | 7               | 84,78  |
|                               | Sugra-39         | 0        | 3              | 3                | 0          | 37       | 0       | 43          | 6               | 86,05  |
|                               | Sultana          | 0        | 0              | 0                | 1          | 0        | 45      | 46          | 1               | 97,83  |
|                               | TOTALS           | 33       | 44             | 48               | 41         | 44       | 50      | 260         |                 |        |
|                               | OMISSIONS        | 2        | 4              | 8                | 2          | 7        | 5       |             |                 |        |
| PRODUCER'S ACCURACY           | 93,94            | 90,91    | 83,33          | 95,12            | 84,09      | 90,00    |         |             |                 |        |
| K                             | 0,1269           | 0,1692   | 0,1846         | 0,1577           | 0,1692     | 0,1923   |         |             |                 |        |
| % Cover                       | 12,69            | 16,92    | 18,46          | 15,77            | 16,92      | 19,23    |         |             |                 |        |
| Total correct classifications | 232              |          |                |                  |            |          |         |             |                 |        |
| Total number of samples       | 260              |          |                |                  |            |          |         |             |                 |        |
| Kappa                         | 0,871            |          |                |                  |            |          |         |             |                 |        |
| Overall accuracy              | 0,892            |          |                |                  |            |          |         |             |                 |        |

Figure 10: Confusion Matrix for Model 6a

### 3.3.2 Model 6b: SVM + Grid Search + PCA

Model 6b (SVM + PCA) performed poorly, achieving an OA of 0.4500 and an MCC of 0.3405. PCA captured maximum variance rather than class separability. In cultivar discrimination, global variance often reflects broad vegetation signals rather than the subtle spectral differences between varieties. PCA likely obscured critical discriminative information, proving that feature extraction is not a guaranteed path to improved accuracy.

|                 |                               | REFERENCE DATA |                |                  |            |          |         | TOTALS | COMMISSIONS | USER'S ACCURACY |
|-----------------|-------------------------------|----------------|----------------|------------------|------------|----------|---------|--------|-------------|-----------------|
|                 |                               | Currants       | Diamond Muscat | Merbein Seedless | Selma Pete | Sugra-39 | Sultana |        |             |                 |
| CLASSIFIED DATA | Currants                      | 27             | 0              | 0                | 2          | 6        | 0       | 35     | 8           | 77,14           |
|                 | Diamond Muscat                | 0              | 20             | 8                | 7          | 6        | 3       | 44     | 24          | 45,45           |
|                 | Merbein Seedless              | 1              | 6              | 16               | 11         | 7        | 5       | 46     | 30          | 34,78           |
|                 | Selma Pete                    | 2              | 5              | 13               | 22         | 3        | 1       | 46     | 24          | 47,83           |
|                 | Sugra-39                      | 4              | 9              | 6                | 6          | 15       | 3       | 43     | 28          | 34,88           |
|                 | Sultana                       | 1              | 7              | 4                | 3          | 14       | 17      | 46     | 29          | 36,96           |
|                 | TOTALS                        | 35             | 47             | 47               | 51         | 51       | 29      | 260    |             |                 |
|                 | OMISSIONS                     | 8              | 27             | 31               | 29         | 36       | 12      |        |             |                 |
|                 | PRODUCER'S ACCURACY           | 77,14          | 42,55          | 34,04            | 43,14      | 29,41    | 58,62   |        |             |                 |
|                 | K                             | 0,1346         | 0,1808         | 0,1808           | 0,1962     | 0,1962   | 0,1115  |        |             |                 |
|                 | % Cover                       | 13,46          | 18,08          | 18,08            | 19,62      | 19,62    | 11,15   |        |             |                 |
|                 | Total correct classifications | 117            |                |                  |            |          |         |        |             |                 |
|                 | Total number of samples       | 260            |                |                  |            |          |         |        |             |                 |
|                 | Kappa                         | 0,339          |                |                  |            |          |         |        |             |                 |
|                 | Overall accuracy              | 0,450          |                |                  |            |          |         |        |             |                 |

Figure 11: Confusion Matrix for Model 6b

### 3.3.3 Model 6c: SVM + Grid Search + Spectral binning

Model 6c (SVM + 5 nm Spectral Binning) achieved an OA of 0.8731 and an MCC of 0.8477. While slightly below the overall accuracy of model 2, its competitive performance confirms that neighbouring narrow bands contain significant redundancy. By aggregating these bands, the model reduced the feature space without sacrificing critical cultivar patterns. This illustrates that spectral binning is a robust dimensionality-reduction strategy that maintains a high signal-to-noise ratio for varietal discrimination.

| CLASSIFIED DATA | REFERENCE DATA                |                |                  |            |          |         |        | COMMISSIONS | USER'S ACCURACY |
|-----------------|-------------------------------|----------------|------------------|------------|----------|---------|--------|-------------|-----------------|
|                 | Currents                      | Diamond Muscat | Merbein Seedless | Selma Pete | Sugra-39 | Sultana | TOTALS |             |                 |
|                 | Currents                      | 32             | 1                | 1          | 1        | 0       | 0      | 35          | 3               |
|                 | Diamond Muscat                | 0              | 40               | 1          | 0        | 3       | 0      | 44          | 4               |
|                 | Merbein Seedless              | 0              | 1                | 37         | 1        | 3       | 4      | 46          | 9               |
|                 | Selma Pete                    | 1              | 0                | 3          | 39       | 2       | 1      | 46          | 7               |
|                 | Sugra-39                      | 1              | 4                | 3          | 0        | 35      | 0      | 43          | 8               |
|                 | Sultana                       | 0              | 0                | 0          | 1        | 1       | 44     | 46          | 2               |
|                 | TOTALS                        | 34             | 46               | 45         | 42       | 44      | 49     | 260         |                 |
|                 | OMISSIONS                     | 2              | 6                | 8          | 3        | 9       | 5      |             |                 |
|                 | PRODUCER'S ACCURACY           | 94,12          | 86,96            | 82,22      | 92,86    | 79,55   | 89,80  |             |                 |
|                 | K                             | 0,1308         | 0,1769           | 0,1731     | 0,1615   | 0,1692  | 0,1885 |             |                 |
|                 | % Cover                       | 13,08          | 17,69            | 17,31      | 16,15    | 16,92   | 18,85  |             |                 |
|                 | Total correct classifications |                |                  |            |          |         |        |             |                 |
|                 | Total number of samples       | 227            |                  |            |          |         |        |             |                 |
|                 |                               | 260            |                  |            |          |         |        |             |                 |
|                 | Kappa                         | 0,847          |                  |            |          |         |        |             |                 |
|                 | Overall accuracy              | 0,873          |                  |            |          |         |        |             |                 |

Figure 12: Confusion Matrix for Model 6c

### 3.4 Temporal transferability

Model 6a (Tuned SVM) was selected for temporal transferability assessment, achieving an OA of 0.2617 and an MCC of 0.1522 on the January dataset. This sharp performance decline is a significant scientific finding, illustrating a lack of model generalisation across different phenological stages. As noted by Loggenberg et al. (2026), spectral signatures are highly sensitive to temporal variability and management practices. These results indicate that the cultivar-specific relationships learned in December were not stable into January, likely due to physiological and canopy shifts over the growing season.

| CLASSIFIED DATA | REFERENCE DATA                |                |                  |            |          |         |        | COMMISSIONS | USER'S ACCURACY |
|-----------------|-------------------------------|----------------|------------------|------------|----------|---------|--------|-------------|-----------------|
|                 | Currents                      | Diamond Muscat | Merbein Seedless | Selma Pete | Sugra-39 | Sultana | TOTALS |             |                 |
|                 | Currents                      | 12             | 0                | 0          | 44       | 0       | 61     | 117         | 105             |
|                 | Diamond Muscat                | 1              | 0                | 1          | 44       | 0       | 98     | 144         | 144             |
|                 | Merbein Seedless              | 1              | 0                | 3          | 17       | 4       | 128    | 153         | 150             |
|                 | Selma Pete                    | 1              | 0                | 0          | 75       | 1       | 76     | 153         | 78              |
|                 | Sugra-39                      | 16             | 0                | 0          | 34       | 0       | 94     | 144         | 144             |
|                 | Sultana                       | 0              | 1                | 3          | 3        | 1       | 133    | 141         | 8               |
|                 | TOTALS                        | 31             | 1                | 7          | 217      | 6       | 590    | 852         |                 |
|                 | OMISSIONS                     | 19             | 1                | 4          | 142      | 6       | 457    |             |                 |
|                 | PRODUCER'S ACCURACY           | 38,71          | 0,00             | 42,86      | 34,56    | 0,00    | 22,54  |             |                 |
|                 | K                             | 0,0364         | 0,0012           | 0,0082     | 0,2547   | 0,0070  | 0,6925 |             |                 |
|                 | % Cover                       | 3,64           | 0,12             | 0,82       | 25,47    | 0,70    | 69,25  |             |                 |
|                 | Total correct classifications |                |                  |            |          |         |        |             |                 |
|                 | Total number of samples       | 223            |                  |            |          |         |        |             |                 |
|                 |                               | 852            |                  |            |          |         |        |             |                 |
|                 | Kappa                         | 0,112          |                  |            |          |         |        |             |                 |
|                 | Overall accuracy              | 0,262          |                  |            |          |         |        |             |                 |

Figure 13: Confusion Matrix for Transferability Test

## 4. Discussion

There is a performance hierarchy for the discrimination of the six raisin cultivars. Model 2 (Baseline SVM) emerged as the best performing model (OA: 0.8923; MCC: 0.8709), demonstrating that for high-dimensional hyperspectral data ( $p=1874$ ;  $n=604$ ), a maximum-margin linear hyperplane provides superior stability over the recursive partitioning of ensemble learners. While Random Forest (Model 1) satisfied the ensemble requirement, its perfect training metrics contrasted with a poor test OA (0.5538), indicating significant overfitting to December-specific noise. This aligns with Gutiérrez et al. (2018), who suggest that linear boundaries are more robust than complex ensembles in scenario where number of predictors ( $p$ ) is much bigger than the number of samples ( $n$ ).

Model 4 (SVM + RFE) performed well and achieved an OA of 0.8769 using only 100 wavebands. This confirms extreme spectral redundancy and demonstrates exceptional feature economy. In contrast, Boruta (Model 3) failed to improve the Random Forest, likely because it retained redundant predictors, while PLS-VIP (Model 5) proved that feature subsets are highly learner-dependent and not universally optimal. Furthermore, the failure of PCA (Model 6b) (OA: 0.4500) illustrates that unsupervised variance maximization does not inherently correspond to class separability, whereas the competitiveness of 5 nm binning (Model 6c) (OA: 0.8731) validates that moderate aggregation can reduce redundancy while maintaining a high signal-to-noise ratio.

The optimisation in model 6a yielded parameters ( $C=1$ , linear kernel) identical to the baseline, confirming that the original specification was already mathematically optimal. However, the most significant finding was the sharp performance decline during the temporal transferability assessment (OA: 0.2617). This indicates that spectral signatures are not temporally stable due to phenological drift the physiological shifts in leaf pigments and structure as the vine matures. As argued by Loggenberg et al. (2026), operational cultivar discrimination requires phenology-aware modelling or multi-temporal training to bridge the spectral gap between different stages of the growing season.

## 5. Conclusion

This study developed and evaluated a machine-learning framework for discriminating six raisin grapevine cultivars from proximal hyperspectral ASD leaf spectra collected in Stellenbosch. It included dataset inspection, atmospheric band removal, class imbalance handling, comparison of an ensemble and non-ensemble learner, learner-specific feature selection, hyperparameter optimisation, feature transformation, and temporal transferability assessment.

The best-performing model was Model 2, the baseline linear SVM using all 1,874 retained wavebands. It achieved the strongest December test performance and therefore became the model selected for refinement in Model 6. Random Forest, which is robust to high dimensionality, performed substantially worse and showed evidence of overfitting. Boruta reduced the RF feature space but did not meaningfully improve RF performance.

Among the reduced-feature models, SVM + RFE was especially important because it retained strong predictive accuracy with only 100 wavebands, indicating that a smaller spectral subset can preserve most of the discriminative information. SVM + PLS-VIP was theoretically justified but less effective for this split. Hyperparameter optimisation of the best model did not improve the baseline SVM, demonstrating that the original linear-kernel,  $C = 1$  configuration was already near-optimal. PCA reduced performance strongly, likely because variance retention did not preserve cultivar-discriminative structure. Spectral binning remained competitive, supporting the idea that moderate aggregation can retain useful spectral patterns while reducing redundancy.

Temporal transferability was poor. The January results dropped sharply relative to December, indicating limited seasonal generalisability and suggesting phenological or temporal spectral drift. The study therefore shows that proximal hyperspectral sensing combined with SVM can discriminate raisin cultivars very effectively within a given time window, but that robust transfer across time remains a major challenge for operational application.

## Appendix A

| Model ID        | Experimental Description                          | OA            | Kappa         | MCC           | Balanced Accuracy | Macro F1      |
|-----------------|---------------------------------------------------|---------------|---------------|---------------|-------------------|---------------|
| <b>Model 1</b>  | Baseline Random Forest                            | 0.5538        | 0.4634        | 0.4641        | 0.5606            | 0.5585        |
| <b>Model 2</b>  | <b>Baseline Linear SVM</b>                        | <b>0.8923</b> | <b>0.8705</b> | <b>0.8709</b> | <b>0.8957</b>     | <b>0.8928</b> |
| <b>Model 3</b>  | Random Forest + Boruta (159 Bands)                | 0.5500        | 0.4590        | 0.4597        | 0.5593            | 0.5576        |
| <b>Model 4</b>  | SVM + Recursive Feature Elimination (100 Bands)   | 0.8769        | 0.8520        | 0.8530        | 0.8839            | 0.8793        |
| <b>Model 5</b>  | SVM + PLS-VIP Selection (100 Bands)               | 0.7154        | 0.6582        | 0.6588        | 0.7202            | 0.7202        |
| <b>Model 6a</b> | Tuned SVM (Grid Search: C=1)                      | 0.8923        | 0.8705        | 0.8709        | 0.8957            | 0.8928        |
| <b>Model 6b</b> | Tuned SVM + PCA Transformation (95% Var)          | 0.4500        | 0.3393        | 0.3405        | 0.4748            | 0.4635        |
| <b>Model 6c</b> | Tuned SVM + 5 nm Spectral Binning                 | 0.8731        | 0.8474        | 0.8477        | 0.8758            | 0.8745        |
| <b>Transfer</b> | <b>Temporal Transfer Test (6a applied to Jan)</b> | <b>0.2617</b> | <b>0.1125</b> | <b>0.1522</b> | <b>0.2311</b>     | <b>0.1615</b> |

## References

- Gutiérrez, S., Fernández-Novales, J., Diago, M. P., & Tardaguila, J. (2018). On-the-go hyperspectral imaging under field conditions and machine learning for the classification of grapevine varieties. *Frontiers in Plant Science*, 9, 1102. <https://doi.org/10.3389/fpls.2018.01102>
- Karakizi, C., Oikonomou, M., & Karantzas, K. (2016). Vineyard detection and vine variety discrimination from very high resolution satellite data. *Remote Sensing*, 8(3), 235. <https://doi.org/10.3390/rs803235>
- Loggenberg, K., Strever, A., & Münch, Z. (2026). Cross-learner spectral subset optimisation: PLS-ensemble feature selection with weighted Borda count for grapevine cultivar discrimination. *Geomatics*, 6(1), 12. <https://doi.org/10.3390/geomatics6010012>
- Mirzaei, M., Marofi, S., Abbasi, M., Solgi, E., Karimi, R., & Verrelst, J. (2019). Scenario-based discrimination of common grapevine varieties using in-field hyperspectral data in the western of Iran. *International Journal of Applied Earth Observation and Geoinformation*, 80, 26–37. <https://doi.org/10.1016/j.jag.2019.04.002>
- Thenkabail, P. S., Smith, R. B., & De Pauw, E. (2000). Hyperspectral vegetation indices and their relationships with agricultural crop characteristics. *Remote Sensing of Environment*, 71(2), 158–182. [https://doi.org/10.1016/S0034-4257\(99\)00067-X](https://doi.org/10.1016/S0034-4257(99)00067-X)